ReRAM Crossbar Compatible Efficient Equality Checker and Half Adder Implementation

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We report on a design and simulation of an adder and an equality checker using Resistive random access memory (ReRAM) compact model implemented in a crossbar architecture based on Memristive-Aided-Logic (MAGIC) family. The designed half-adder computes the result in six execution steps, which is the minimum reported so far, using MAGIC in crossbar architecture. The equality checker demonstrated here, is designed to perform parallel executions, such that the number of execution steps is independent of the number of input bits.

As machine learning and AI gain widespread adoption, the traditional von Neumann architecture encounters a significant bottleneck in terms of computational performance and energy efficiency. This is often attributed to the "memory wall" issue arising from the physical separation of storage and computational units in von-Neumann architecture. Emerging non-volatile memory technologies such as Phase Change Memory (PCM), Ferroelectric Field-effect Transistor (FeFET), Magnetic Random Access Memory (MRAM), and Resistive Random Access Memory (ReRAM) have shown the potential to overcome the "memory wall" issue by offering in-memory computation capability [1]. In-memory computing paradigm performs the computation in the storage unit itself, thus eliminating the need for data movement, consequently offering a computation performance boost. ReRAM technology, due to its compact two-terminal structure, CMOS compatibility, high switching speed and low energy consumption, is a strong contender for in-memory computing-based systems [1]. Furthermore, ReRAM devices, when connected in a specific manner called a crossbar array, enable parallel processing of large amounts of data. This work presents a ReRAM crossbar-compatible implementation of two widely used logic circuits, a 2-bit half adder and an 8-bit similarity checker. The proposed designs are implemented in SPICE and validated using physics-based ReRAM compact models [2]. The implementation of the presented logic gate is based on the Memristor Aided Logic (MAGIC) family, as it offers several advantages compared to other logic families. For example, MAGIC requires only one voltage source and uses dedicated ReRAM for output, thus separating input and output [3]. The specific novel components of the presented work are: 1) the designed 2-bit half-adder computes the result in six execution steps, which is the minimum reported so far [4]; and 2) the similarity checker, demonstrated here, exploits the parallel computation capability of a ReRAM crossbar resulting in the number of execution steps (six steps) independent of the number of input bits.

Fig. 1(a) shows a ReRAM-based 2-input NOR gate implementation. The input memristors (shown in red) are connected in parallel, and an additional memristor (shown in green) is used to store the output. Initially, the output memristor is initialised to the low resistance state. During the logic execution step, voltage V_{in} is applied at the input. For the case when both input memristors are in HRS, the voltage drop across the output memristor will be less than its threshold voltage; consequently, its logical state remains unchanged. For all other input combinations, the voltage drop across the output memristor exceeds its threshold and thus changing its logical state. The corresponding truth table for the 2-input NOR gate is shown in Fig. 1(b). The series connection of input and output memristors with opposite polarity enables the logical NOT operation, as shown in Fig. 1(c).

The half-adder circuit demonstrated here, shown in Fig. 1(d), takes two inputs, P and Q and gives two outputs, Sum and Carry. The value of SUM is P XOR Q, and the value of the Carry is P AND Q. Here in this work, we implement NOR gates along the columns instead of rows. In the first step, we parallelly calculate the NOT of P and Q (denoted by P^* and Q^*) and store it on the memristors to the right of P and Q in the second column. In the second step, we take the NOR of P and Q, and also we take the NOR of P^* and Q^*, which gives us P AND Q. Next, we compute the following in discrete

steps - P NOR Q, P NOR (P NOR Q) which we shall call X and we calculate Q NOR (P NOR Q) which we shall call Y. We then calculate X NOR Y, which essentially gives P XNOR Q. We then invert this using a NOT gate to get P XOR Q which is the SUM. Hence, in 6 execution steps, we are able to get the values of SUM and CARRY.

An equality checker is a digital circuit which accepts two n-bit keys and checks them for equality. The circuit returns "1" if both keys are equal or "0" otherwise. The two keys are stored vertically in the first and the second column, respectively, as shown in Fig. 1(e). In the proposed design, all the operations are done parallelly on corresponding bits of the two keys. Let A_i and B_i represent the ith bit of the keys. In the first step, we take A_i NOR B_i, followed by A_i NOR (A_i NOR B_i), labelled as X_i and B_i NOR (A_i NOR B_i), labelled as Y_i, in the subsequent steps. We then calculate X_i NOR Y_i, which essentially gives A_i XNOR B_i. Next, we invert the obtained bits using a NOT gate and store them in the last column (Step 5). Finally, we perform the NOR operation to get the result.

In summary, our works pave the way towards compact and energy-efficient in-memory computing-based digital logic circuit hardware, which is essential for data-intensive applications.

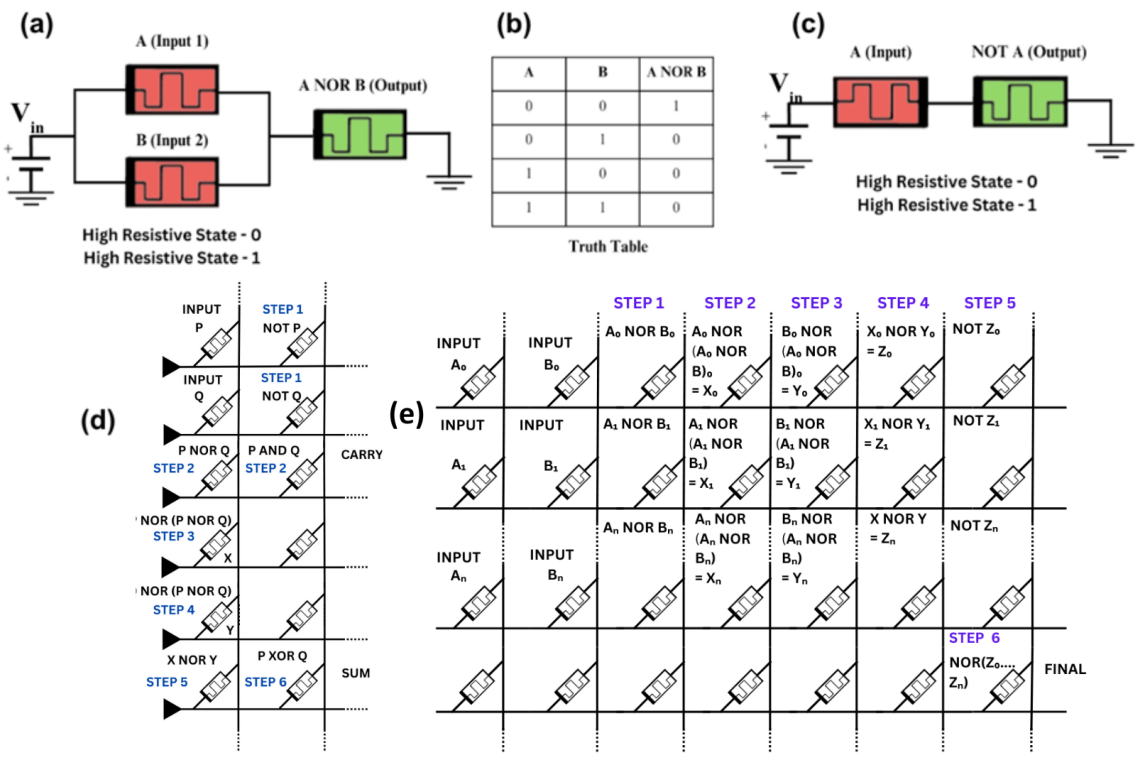

Figure 1. (a) 2-bit NOR gate implementation; (b) NOR gate truth table; (c) NOT gate implementation; and Crossbar compatible (d) 2-bit Half adder & (e) n-bit equality checker implementation.

References:

[1] Yang, J., Strukov, D. & Stewart, D., *Nat. Nanotech.* **8**, 13 (2013).

[2] Jiang, Z., Wu, Y., Yu, S., Yang, L., Song, K., Karim, Z., & Wong, H. S. P., *IEEE Trans. Electron Devices*, **63**(5), 1884 (2016).

[3] Kvatinsky S., Belousov D., Liman S., Satat G., Wald N., Friedman E. & Kolodny A., *IEEE Trans. Circuits Syst. II: Express Br.*, **61**(11), 895 (2014)

[4] Thangkhiew, P.L., Gharpinde R., Yadav D. N., Datta K. & Sengupta I., *IEEE Reg. 10 Annu. Int. Conf. Proc./TENCON,* 207 (2017)